

2.6.30

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

If the area of a triangle is 35 square units with vertices $(2, 6)$, $(5, -4)$ and $(k, 4)$, then find k .

Solution: Let $\vec{A} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$, $\vec{B} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$, $\vec{C} = \begin{pmatrix} k \\ 4 \end{pmatrix}$

$$\text{Vectors: } \vec{B} - \vec{A} = \begin{pmatrix} 5-2 \\ -4-6 \end{pmatrix} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \quad \vec{C} - \vec{A} = \begin{pmatrix} k-2 \\ 4-6 \end{pmatrix} = \begin{pmatrix} k-2 \\ -2 \end{pmatrix}$$

$$\text{The area of the triangle is Area} = \frac{1}{2} |(\vec{B} - \vec{A}) \times (\vec{C} - \vec{A})|$$

$$35 = \frac{1}{2} \left| \begin{pmatrix} 3 \\ -10 \end{pmatrix} \times \begin{pmatrix} k-2 \\ -2 \end{pmatrix} \right|$$

So,

$$\begin{aligned} 35 &= \frac{1}{2} |3(-2) - (-10)(k-2)| \\ &= \frac{1}{2} |-6 + 10k - 20| \\ &= \frac{1}{2} |10k - 26| \end{aligned}$$

$$\text{Thus, } |10k - 26| = 70$$

$$10k - 26 = 70 \quad \Rightarrow \quad k = \frac{48}{5}$$

$$10k - 26 = -70 \quad \Rightarrow \quad k = -\frac{22}{5}$$

$$\text{Therefore, } k = \frac{48}{5} \quad \text{or} \quad k = -\frac{22}{5}$$

Triangles with vertices $(2,6)$, $(5,-4)$, and $(k,4)$ for $k=-4.4$ and $k=9.6$

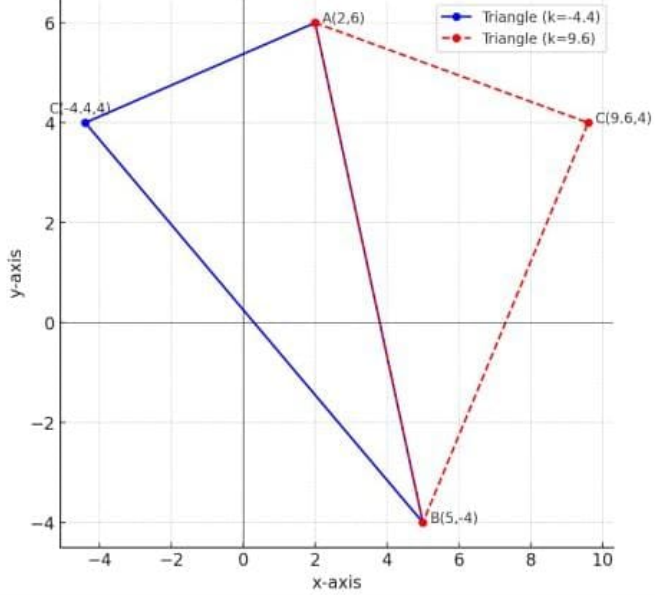


Fig. 0.1